CAPSTONE PROJECT PROPOSAL

Project Title: TBD

Course: SOFT290 | Instructor: Milton Cruz

Submitted: April 12, 2026

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1. Project Title and Team Members

Project Title: TBD — to be finalized by the team prior to submission.

Team Member	Track / Role	Responsibilities
Leena	BDA — Team Lead	Team Lead and project manager. Full database ownership: data gathering, cleaning, schema design, ingestion, and all stored procedures. KPI definitions, EDA, and data visualization dashboards (prototyped in Power BI, integrated into the app via React charting libraries). Primary .NET 8 backend developer. Backend scope and task coordination with SD team. Pull request reviews and main branch management.
Eli	SD — Dev 1	Lead UI/UX designer. Primary React Native front end developer — establishes web UI screens, navigation, layout, globe/map integration, and core UI components. Backend endpoint support on clearly defined, scoped tasks once architecture is established.
Landin	SD — Dev 2	React Native front end development with a focus on mobile design. Adapts established web UI screens for mobile — optimizing layouts, touch targets, and navigation patterns for a native mobile experience. Owns implementation of designated UI screens and components across both platforms. Backend endpoint support on clearly defined, scoped tasks once architecture is established.

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2. Project Description

Summary

This project is a cross-regional music discovery platform that analyzes how popular music overlaps across countries and how those patterns change over time. Using real Spotify chart data spanning 70+ countries and multiple years, the platform surfaces the 'Discovery Gap' — globally trending songs and artists that have not yet crossed over into a user's local market.

The application is built around two core experiences: an interactive Discovery Globe that lets users explore regional music data on their own terms, and a Global Overlap Dashboard that provides a direct, data-driven answer to the project's central question. Both are accessible from a Welcome Screen that orients the user and offers entry points into either experience.

The Problem

Music discovery is regionally siloed. Even though streaming platforms make all music globally accessible, chart algorithms and recommendation systems reinforce local listening habits. A song can be charting in the Top 10 across a dozen countries while remaining completely unknown in another. There is currently no consumer-facing tool that clearly visualizes this gap or helps users identify music they are likely to enjoy based on cross-regional overlap.

Target Users and Use Cases

- Music listeners interested in discovering globally trending artists and tracks before they break locally
- Users curious about cultural differences in music taste across regions
- Data-curious users who want to explore how music spreads geographically over time

Core Use Cases

- Explore regional music relationships by interacting with a globe — selecting countries to surface chart data, comparisons, and hidden gems
- Compare music chart overlap between two countries — shared songs, unique songs, and overlap percentage
- Identify songs popular in surrounding regions but absent from a selected country's charts
- View a direct analytical answer to the Discovery Gap question through trend visualizations and KPI dashboards
- Track how global music overlap changes over time

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3. Technology Stack

Layer	Technology	Justification
Project Management	GitHub Projects — https://github.com/users/lkomenski/projects/6	All tasks, deadlines, and assignments are tracked in a GitHub Project board linked to the repository. Every deliverable in this proposal corresponds to a GitHub Issue assigned to a team member. The Team Lead manages the board, reviews pull requests, and enforces branch protection on main.
Database	Microsoft SQL Server (MSSQL)	Robust relational database suited for large datasets (15M+ rows). Stored procedures handle all heavy computation — overlap percentages, trend velocity, Hidden Gem logic

Layer	Technology	Justification
		— keeping processing server-side and efficient.
Backend	.NET 8 (C#)	Professional-grade API layer. Decoupled from the front end for clean separation of concerns and platform flexibility. Repository Pattern with DTOs ensures a clean data handshake between the database and UI layers.
Frontend	React Native + React Native Web	Single codebase targets both mobile and web platforms simultaneously. Reduces development time and ensures feature parity across platforms.
Data Visualization	React charting libraries — to be finalized during development (candidates: Recharts or Victory Native)	Power BI cannot be fully integrated into a deployed web/mobile application due to licensing constraints. React charting libraries replicate the required dashboards directly within the app, ensuring all BDA deliverables are visible within the product itself. The final library selection will be made once development begins — candidates will be evaluated for friction, React Native compatibility, and rendering performance before committing to either.
External APIs	Interactive globe/map API — to be finalized during development (candidates: Mapbox, Google Maps, react-globe.gl, or similar). It is too early to commit before hands-on testing determines which introduces the least friction in a React Native environment.	An external API will power the interactive globe visualization, providing geographic rendering and country selection capabilities. Candidates will be evaluated in Week 2 for React Native compatibility, mobile performance, and documentation quality. All selected APIs must have free tiers appropriate for an academic project.
Data Sources	Kaggle: Spotify Historical Charts (2017-2021) + Top Spotify Songs in 73 Countries (2023-2025)	Two complementary datasets providing broad geographic coverage (73 countries) and historical depth. Combined size: ~4GB.
Version Control	GitHub	Industry-standard source control. Branch protection on main, pull request reviews by Team Lead, and GitHub Issues used for all task tracking throughout the project.
BDA Prototyping	Power BI (design/prototyping only)	Used internally by the BDA team to design, validate, and prototype dashboard layouts before implementation in React. Not included in the final deployed product.

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4. Features and Functionality

Core MVP Features

The application is structured around three views. The Welcome Screen is the entry point. The Discovery Globe is the primary user-driven exploration experience. The Global Overlap Dashboard is the direct analytical answer to the project's core question.

Feature	Description
Welcome Screen	
Entry + Navigation	Project purpose, data disclaimer noting the 22-month gap between datasets, and navigation to both core views. Users can choose to explore freely via the globe or begin with a set of preset filters — both paths lead to the Discovery Globe.
Discovery Globe	
Interactive Globe	The primary exploration interface. An interactive globe powered by an external map API allows users to click on any country to surface a curated dataset. The layout is a single rich page — country data, comparisons, and song details all surface contextually within the same view (via side panels, popups, or similar — final layout determined during design phase).
Country Profile	Selecting a country on the globe surfaces that country's chart data, including, but not limited to: shared vs. unique song distribution, top shared songs, and top region-specific songs.
Country Comparison View, Country Comparison Mode	Selecting up to three countries triggers a comparison view, including but not limited to: overlap count, shared percentage, and lists of shared and unique songs for each country.
Hidden Gems Page	Surfaced contextually from the Country Profile or Comparison view. Shows songs (with 30-second sound previews) trending in surrounding or peer countries that are absent from the selected country's charts. Filterable by date range and minimum country count threshold.
Filters Page	Displays the interactive globe alongside a persistent filter panel, allowing users to narrow the data they are exploring. Filters surface more specific chart data — for example, by date range, country, chart type, or minimum country count threshold — enabling users to customize their exploration and discover more targeted results without leaving the globe view.
Global Overlap Dashboard	
Trend Visualizations + KPIs	The direct analytical answer to the project's core question: which globally trending artists and songs are missing from local markets, and how do these patterns change over time? Four headline KPIs are displayed as prominent summary figures at the top of the dashboard: (1) Global Overlap Rate — the percentage of all charting songs that appeared in 2 or more countries; (2) Average Discovery Gap — the average number of days between a song's first chart appearance in any country and its appearance in a second country; (3) Most Globally Isolated Region — the country with the highest percentage of locally unique charting songs; and (4) Peak Cross-Regional Reach — the maximum number of countries a single song simultaneously charted in, displayed alongside the song title and artist name as a Vinyl card.

Feature	Description
	Below the KPIs, trend line charts and global distribution metrics show how overlap and discovery patterns shift across the dataset's two periods (2017–2021 and 2023–2025). All KPIs are calculated via stored procedures and are filterable by date range.
App-Wide	
Dark Mode First	The entire application is designed and built dark mode first. Dark mode is the default and primary visual experience across all screens, components, and platforms.

Potential Future Enhancements

The following features are not planned for the current project timeline but represent natural extensions of the platform if time and resources allow after all MVP features are complete and stable:

- Light mode toggle
- Audio previews for tracks within the app
- Timeline view tracing a song's geographic spread — for example, which country it charted in first
- Audio feature and mood comparison using the dataset's danceability, tempo, valence, and energy columns — a corresponding reach goal would be to source the missing audio feature data (danceability, tempo, valence, energy, etc.) from external APIs such as the Spotify API, since only one of the two datasets includes this data. Completing the dataset via external sourcing would enable the full audio feature experience across the entire historical range.
- Accessibility features

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5. Milestones and Timeline

The following table provides a milestone-by-milestone breakdown of all project deliverables, assigned owners, and corresponding GitHub Issue labels. Week ranges are used throughout; specific target dates are noted only for major deliverable handoffs. Documentation tasks are woven into each milestone rather than deferred to the end. Any changes to the timeline will be documented in the GitHub Project board and reflected in updated GitHub Issues.

The team holds twice-weekly standing check-ins to review progress, surface blockers, and align on priorities. An additional standing check-in is scheduled every other week as an optional buffer meeting, used only when needed.

Owner color coding: Leena / BDA (green) | Eli / SD Dev 1 (blue) | Landin / SD Dev 2 (orange) | All (purple)

GitHub Issue Labels column is for internal project tracking — each label corresponds to a GitHub Issue on the project board.

Timeline	Task	Owner	GitHub Issue Label
Note: Week 2 of the project corresponds to Week 3 of the quarter. Any changes to the timeline will be documented in the GitHub Project board and reflected in updated GitHub Issues.			
WEEK 2 — Design Phase | 4/5 – 4/11			
By 4/12	Finalize and submit agreed-upon proposal draft	**All**	[PROPOSAL] Finalize and submit capstone proposal
Week of 4/7	Define and document fully normalized database schema (all tables, PKs, FKs, relationships) + accompanying ADR justifying all major design decisions	**Leena**	[DATA] Database schema + ADR
Week of 4/7	Define all KPIs: overlap %, discovery gap score, trend velocity — document formulas and calculation logic	**Leena**	[BDA] KPI definitions document
Week of 4/7	Produce detailed UI sketches for all views including panels, popups, and secondary UI states — sketches serve as the build specification for the SD team	**Eli**	[UI] Detailed UI sketches — all views + secondary states
By 4/10	Team check-in: agree on finalized UI designs, font choices, and dark mode color palette	**All**	[UI] Design system agreement — fonts, colors, dark mode base
By 4/11	Research globe/map API and React charting library candidates — document finalists with justification + ADR	**Eli**	[TECH] React library research + ADR — globe + charting
WEEK 3 — Front End Build & Data Prep | 4/12 – 4/18			
Leena unavailable 4/11–4/16. Leena resumes 4/17.			
Eli unavailable 4/17–4/19. Eli resumes 4/20.			
Week of 4/12	Set up React Native project scaffold — navigation structure, folder architecture, and dark mode base stylesheet + component architecture document	**Eli**	[FRONTEND] Project scaffold + navigation skeleton
Week of 4/12	Build core UI shells — Discovery Globe page shell (globe renders, country selection works, panel/popup structure in place with dummy data), Global Overlap Dashboard page shell (layout, chart placeholders, KPI summary areas with no live	**Eli**	[FRONTEND] Core UI shells — Globe, Dashboard, Vinyl card

Timeline	Task	Owner	GitHub Issue Label
	data), and Vinyl card component (reusable track/album display, dark mode styled, with variations: explicit tag, rank badge, trend indicator)		
Week of 4/12	Build Welcome Screen shell — entry copy, data disclaimer, navigation to both core views, filter entry point	**Landin**	[FRONTEND] Welcome Screen shell
Week of 4/12	Build discrete UI components as identified from Eli's sketches — forms, filter panels, nav bars, search inputs, and other self-contained elements	**Landin**	[FRONTEND] Discrete UI components — per sketch spec
Week of 4/17	Dataset 1 and Dataset 2 cleaning begun — Power BI cleaning pass on both datasets	**Leena**	[DATA] Dataset cleaning — both datasets
Week of 4/17	Define stored procedure signatures for all core queries (inputs, outputs, logic outline) + data cleaning summary document	**Leena**	[DATA] SP design + cleaning summary docs
WEEK 4 — Database Completion + Backend Skeleton | 4/19 – 4/25			
Eli unavailable through 4/19. Eli resumes 4/20. Full team at capacity from 4/20 onward.			
4/19	Full database created, all tables populated, DB creation script finalized and committed to repo	**Leena**	[DATA] Full database creation — 4/19 target
4/19	Both datasets fully cleaned and ingested via BULK INSERT — all rows validated	**Leena**	[DATA] Dataset ingestion — BULK INSERT both datasets
4/19	All lookup tables populated: Countries, ChartType, Artist, Song, ArtistSong, tbd	**Leena**	[DATA] Lookup table population
Week of 4/20	Set up .NET 8 project structure (solution, folder architecture, repository pattern scaffold) + ADR documenting architectural decisions	**Leena**	[BACKEND] .NET 8 project setup + architecture ADR
Week of 4/20	Write all core stored procedures: Overlap %, Country Comparison, Global Trend, Country Profile, etc. — documented inline as written	**Leena**	[DATA] Core stored procedures — all MVP queries
Week of 4/20	Set up .NET API routes, controller scaffolding, and DTO	**Leena**	[BACKEND] API routes + controller scaffolding + DTOs

Timeline	Task	Owner	GitHub Issue Label
	definitions for all planned endpoints		
Week of 4/20	Refine Discovery Globe page — globe interaction logic, hover states, country click, selected state highlight, and transition animations	**Eli / Landin**	[FRONTEND] Globe interaction logic + animations
Week of 4/20	Build chart placeholder components with hardcoded dummy data — Global Overlap trend, Country Profile breakdown	**Eli / Landin**	[FRONTEND] Chart placeholders with IMCdummy data
Week of 4/20	Implement Country Comparison endpoint — calls SP, maps to DTO, returns JSON	**Eli**	[BACKEND] Country Comparison endpoint
Week of 4/20	Implement Country Profile endpoint — calls SP, maps to DTO, returns JSON	**Eli**	[BACKEND] Country Profile endpoint
Week of 4/20	Build additional discrete UI components identified during Eli's build process — loading states, empty states, error states	**Landin**	[FRONTEND] Additional discrete UI components
Week of 4/20	Mobile adaptation begins: adapt Welcome Screen and any completed shells for mobile layout	**Landin**	[FRONTEND] Mobile adaptation — Welcome Screen
WEEK 5 — Remaining Endpoints & Data Visualization | 4/26 – 5/2			
Week of 4/26	Implement Hidden Songs Explorer endpoint — calls SP, maps to DTO, returns JSON	**Leena**	[BACKEND] Hidden Songs endpoint
Week of 4/26	Implement Global Overlap Trend endpoint — calls SP, maps to DTO, returns JSON	**Leena**	[BACKEND] Global Overlap Trend endpoint
Week of 4/26	Prototype all dashboard and chart designs in Power BI — reference designs for React implementation	**Leena**	[BDA] Power BI dashboard prototypes — all core views
Week of 4/26	Implement React chart components: Global Overlap trend line chart + Country Profile breakdown chart — documented inline as built	**Leena**	[FRONTEND] Chart components — Overlap trend + Profile
Week of 4/26	Implement Global Overlap Trend endpoint — calls SP, maps to DTO, returns JSON	**Eli**	[BACKEND] Global Overlap Trend endpoint (SD)

Timeline	Task	Owner	GitHub Issue Label
Week of 4/26	Wire Discovery Globe page to live Country Profile and Country Comparison endpoints — replace dummy data with real API responses	**Eli / Landin**	[FRONTEND] Globe page — live data hookup
Week of 4/26	Wire Global Overlap Dashboard to live trend endpoint — chart components receive real data	**Leena**	[FRONTEND] Dashboard — live data hookup
Week of 4/26	Mobile adaptation: adapt Discovery Globe page for mobile — touch interactions, panel/popup behavior on small screens	**Landin**	[FRONTEND] Mobile adaptation — Discovery Globe page
Week of 4/26	Build any additional discrete UI components identified during Week 4-5 build process	**Landin**	[FRONTEND] Additional discrete UI components — Week 5
WEEK 6 — Integration Debugging & Polish | 5/3 – 5/9			
This week is reserved primarily as a debugging buffer for issues that surface when real data is connected. If integration is stable, time shifts to polish.			
Week of 5/3	Debug and resolve any issues surfaced from live data hookup — SP output mismatches, DTO shape errors, chart rendering issues	**All**	[QA] Integration debugging — live data issues
Week of 5/3	Cross-platform rendering verification — confirm the application renders and functions correctly across iOS, Android, Windows desktop, and web browser targets. Validate layout, navigation, touch targets, and feature parity across all environments. Target: 2–3 days; worst-case buffer of one full week absorbed within this milestone.	**Eli**	[QA] Cross-platform rendering verification
Week of 5/3	Web UI polish pass: verify all screens render correctly on web — layout, spacing, typography, and interactions	**Eli / Landin**	[FRONTEND] Web UI polish — all screens
Week of 5/3	Mobile adaptation: adapt Global Overlap Dashboard for mobile	**Eli / Landin**	[FRONTEND] Mobile adaptation — Dashboard page
Week of 5/3	Full dark mode audit across all screens on both web and mobile	**Eli / Landin**	[UI] Dark mode audit — web + mobile

Timeline	Task	Owner	GitHub Issue Label
By 5/9	Design review: all screens match agreed sketches on both platforms — document any intentional deviations	**All**	[DOCS] Design review sign-off
WEEK 7 — Integration Sprint | 5/10 – 5/16			
Week of 5/10	Connect Hidden Songs Explorer to live API endpoint within Discovery Globe page	**Eli**	[INTEGRATION] Hidden Songs — API wired to UI
Week of 5/10	End-to-end flow verification: Welcome Screen → globe → country profile → comparison → hidden songs → dashboard	**All**	[QA] End-to-end flow verification
Week of 5/10	Connect any remaining endpoints not yet wired to UI	**Leena**	[INTEGRATION] Remaining endpoint connections
By 5/16	End-to-end integration test: full user flow verified on both web and mobile. Document results.	**All**	[QA] End-to-end integration test + results doc
WEEK 8 — Testing & QA | 5/17 – 5/23			
Week of 5/17	API endpoint testing — response accuracy, error handling, and edge cases across all routes (individual test cases tracked as GitHub Issues)	**Eli**	[QA] API endpoint testing
Week of 5/17	UI testing — all screens on web: layout, navigation, interactions, and edge case behavior	**Landin**	[QA] UI testing — web
Week of 5/17	Stored procedure validation — verify overlap %, Hidden Gems/other logic, and trend calculations against known data (individual cases tracked as GitHub Issues)	**Leena**	[QA] Stored procedure validation
Week of 5/17	UI testing — all screens on mobile: touch targets, layout reflow, navigation, and mobile-specific edge cases	**Landin**	[QA] UI testing — mobile
Week of 5/17	Performance testing — large dataset query response times and chart render times under realistic load	**All**	[QA] Performance testing
Week of 5/17	Bug fixes — all issues surfaced during testing addressed and verified (individual bugs tracked as GitHub Issues)	**All**	[QA] Bug fixes

Timeline	Task	Owner	GitHub Issue Label
By 5/23	QA findings log — document all bugs found, fixes applied, and any known remaining limitations	**All**	[QA] QA findings log
WEEK 9 — Deployment & Final Presentation | 5/24 – 5/31			
Week of 5/24	Research and agree on cloud deployment platform as a team — document platform decision and setup plan	**All**	[DEPLOY] Cloud platform selection + plan
Week of 5/24	Deploy .NET 8 backend API to cloud environment	**Leena**	[DEPLOY] Backend API deployment
Week of 5/24	Deploy React Native web build to cloud environment	**Eli**	[DEPLOY] Frontend web deployment
Week of 5/24	Full production smoke test — all features validated in live environment	**All**	[DEPLOY] Production smoke test
Week of 5/24	Write ADR: Deployment decisions — platform choice, configuration, and lessons learned	**Leena**	[DOCS] ADR — Deployment decisions
By 5/30	Write README and deployment guide: project overview, setup instructions, environment variables, deployment steps, and API documentation	**All**	[DOCS] README + deployment guide
WEEK 9 (CONTINUED) — Final Presentation | 5/31			
Week of 5/31	Final documentation audit: verify all major decisions have ADRs, all components are documented, README is complete	**All**	[DOCS] Final documentation audit
Week of 5/31	Prepare demo script and walkthrough: Welcome Screen → globe → country profile → comparison → hidden songs → dashboard	**All**	[DOCS] Demo script and walkthrough
Week of 5/31	Final code review and cleanup — remove dead code, ensure inline comments are complete	**All**	[QA] Final code review + cleanup
By 5/30	Presentation rehearsal — all team members walk through their portion	**All**	[FINAL] Presentation rehearsal
5/31	Final presentation delivered	**All**	[FINAL] Capstone presentation

6. Challenges and Risks

Challenge / Risk	Mitigation Strategy
Dataset size (~4GB combined, 15M+ rows)	All heavy computation (overlap %, Hidden Gems logic, trend aggregation) lives in MSSQL Stored Procedures rather than application code. Only cleaned, aggregated results are passed to the API layer, keeping response times fast and the application layer lightweight.
Power BI cannot be deployed within a web or mobile application	Power BI is used internally as a prototyping and design tool only. All dashboards and data visualizations are implemented directly in the React Native front end using React charting libraries, ensuring all BDA deliverables are fully visible and functional within the deployed product.
22-month gap in chart data (Dec 2021 – Oct 2023)	The data gap is documented transparently on the Welcome Screen as a known data limitation. Time-series visualizations will clearly mark this gap to avoid misleading trend interpretations. The limitation is also noted in all relevant documentation.
Varying experience levels across team members	Tasks are assigned in proportion to each team member's verified skill level and areas of strength. Self-contained, clearly defined tasks are assigned to ensure no team member's work blocks another's. Backend support tasks are scoped and documented before assignment so expectations are unambiguous.
Data hookup debugging when live endpoints replace dummy data	Week 6 is reserved primarily as an integration debugging buffer. Rather than treating data hookup as a single integration sprint event, live data is connected incrementally starting in Week 5 so issues surface early. Week 6 exists specifically to absorb unexpected debugging time before the formal QA phase.
Team member availability gaps during critical early weeks	Known unavailability windows are accounted for in the timeline — no tasks are assigned to unavailable team members during those periods. The SD team's front end work is structured to proceed independently during the Team Lead's absence, and vice versa.
External API / globe library compatibility with React Native	Library candidates are evaluated in Week 2 specifically for React Native and mobile compatibility before any code is written. Libraries with dedicated React Native documentation and active maintenance will be prioritized. Performance testing is included in the Week 8 QA milestone.
Merge conflicts and branch management across three developers	Branch protection rules on main require pull request review and approval by the Team Lead before merging. Feature branches are scoped to individual GitHub Issues. Twice-weekly check-ins keep all team members aligned on active branches and in-progress work.
Cloud deployment (no prior experience across the team)	Deployment is new territory for all three team members. It is scheduled for Week 9, with the final presentation on 5/31. The team will research and select a platform together, and the deployment process will be documented step-by-step as a learning record and for the final README.

Challenge / Risk	Mitigation Strategy
Protecting intellectual property and documenting the work process	All work will be tracked through properly formatted GitHub Issues and pull requests with clear dates, descriptive titles, and detailed notes explaining what was done and why. Commit messages will follow a consistent convention. This documentation serves both as a clear record of who contributed what and when, and as protection of the team's intellectual property throughout the project lifecycle. The Team Lead enforces this standard during all pull request reviews.
Managing a large local dataset and sharing the database across multiple developers	This is the first time the Team Lead has worked with a dataset of this scale locally (~4GB, 15M+ rows), and sharing a live database across multiple team members introduces risk around consistency and performance. Mitigation strategies include: monitoring query performance from the start and adding indexes where slowdowns are detected; reviewing and adjusting stored procedures and schema design iteratively as processing speeds are observed; documenting the database setup process step-by-step so all team members can replicate the environment; and agreeing on a shared approach to database access (such as a shared SQL Server instance or clearly defined restore and seed scripts) before development begins.

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7. Evaluation Criteria

How We Measure Success

Functional Criteria

- All core MVP features are implemented and working as intended with no blocking bugs
- The application clearly and obviously answers the core question: 'Which globally trending artists and songs are missing from my local market, and how do these patterns change over time?'
- All stored procedures produce accurate results and are validated against known data during QA
- All API endpoints return correct, well-structured responses and handle edge cases gracefully
- Data visualizations accurately reflect the underlying database query results
- The application does not break when used outside of intended flows — edge cases are anticipated and handled
- Cross-platform visual and functional consistency — the application renders correctly and behaves equivalently across iOS, Android, Windows desktop, and web browser targets. Layout, navigation, touch interactions, and all features are verified on each platform; the experience is not degraded or broken on any supported environment.

BDA Track Criteria

- Database schema is fully normalized, well-documented, and logically sound
- All KPIs are clearly defined, consistently calculated via stored procedures, and visible within the application
- Dashboards and data visualizations are integrated directly into the app UI and match prototyped Power BI designs
- All analytical logic uses statistical SQL (window functions, aggregates) — no machine learning models

SD Track Criteria

- React Native application runs correctly on both mobile and web from a single codebase
- The application is dark mode first throughout — consistent across all screens and both platforms
- .NET 8 API is properly decoupled from the front end using Repository Pattern and DTOs
- Application is successfully deployed to a cloud environment and fully demonstrable in production

Documentation Criteria

- All major architectural decisions have a corresponding Architecture Decision Record (ADR) committed to the repo
- README covers project overview, local setup, environment configuration, and API usage
- All stored procedures, endpoints, and chart components are individually documented
- Data cleaning decisions and known limitations are documented and visible to the end user where relevant

What a Complete Project Looks Like

- **All agreed-upon MVP features work correctly across mobile and web with no blocking bugs**
- **The application is deployed, publicly accessible, and fully demonstrable in a live environment**
- **Data layer, API layer, and UI layer are fully integrated end-to-end**
- **The UI is intuitive, visually polished, and dark mode first throughout**
- **Documentation is complete: README, ADRs, endpoint docs, component docs, and in-app data disclaimer**
- **The project clearly and obviously answers the core problem it set out to solve**